

NETWORK DEVICE

TEST

Date |

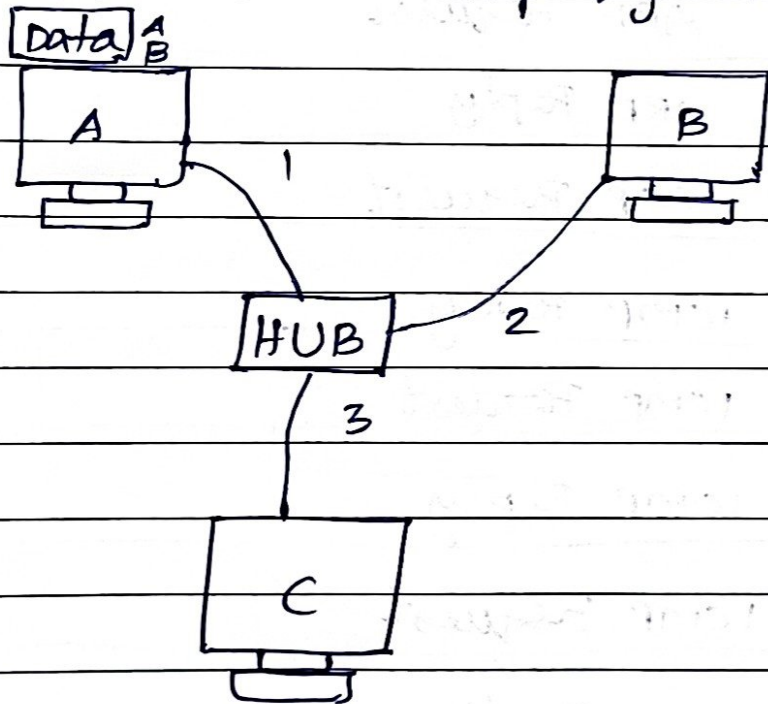
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1) Explain behaviour of hub device?

HUB :

- * It is used to connect the multiple network device.
- * It is a broadcast device.

Let us see in step-by-step :-



* we have three pc's, name as A, B, and C and port no is A-1, B-2, and C-3

* First of all, the data is send to PC A to PC B.

first, you have to

- * Send the data to hub, once the hub is received the data.

- * the data is can be broadcast to all pc.

- * send for pc B and pc C

- * pc B receives the data and pc C receive and automatically drop the data.

- * Main think, once the data is shared to the hub, it can be broadcast and collision.

- * Because, of data loss and heavy traffic.

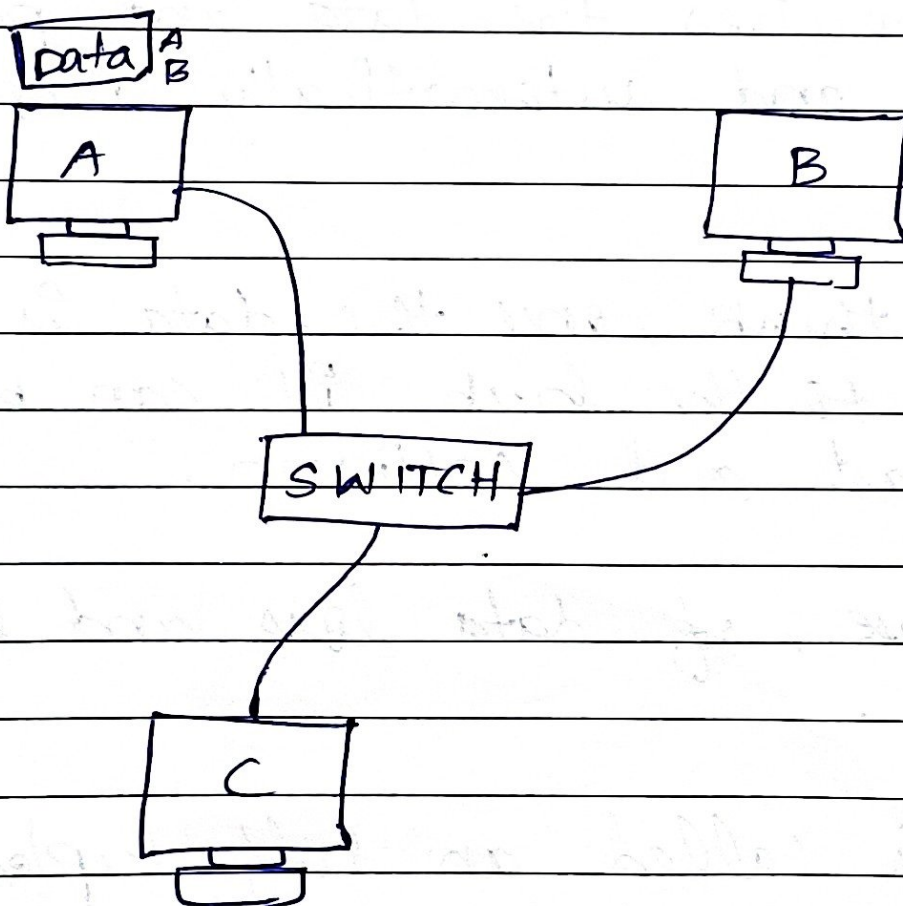
- * It is called an half duplex.

- * The hub behaviour is once you share the data from one pc to another pc, using the hub, the data can be broadcasted.

2) Explain behaviour of switch device?

SWITCH :-

- * It is use to connect multiple network device.
- * It contains mac table.
- * It will act as broadcast + unicast.



* shared the data from one pc. to another pc.

* we have three pc = A, B, C

* First, you have to assign a ip address of A and B pc. ~~and mac addresses in~~

* ~~mac~~ A = 50.0.0.10 and B = 50.0.0.20

* we have to send the echo data from pc B with the help of switch.

* The switch behaves, once the data is received and note the port number and MAC address on the MAC table is known as Address learning.

* If first time, you will send the data and stores on the mac table and broadcast to all pc.

* Then return from pc B to pc A, it will comes in switch, it will noted the pc B port number and mac address and check it whether the destination is on the Mac table and directly send to the

destination pc.

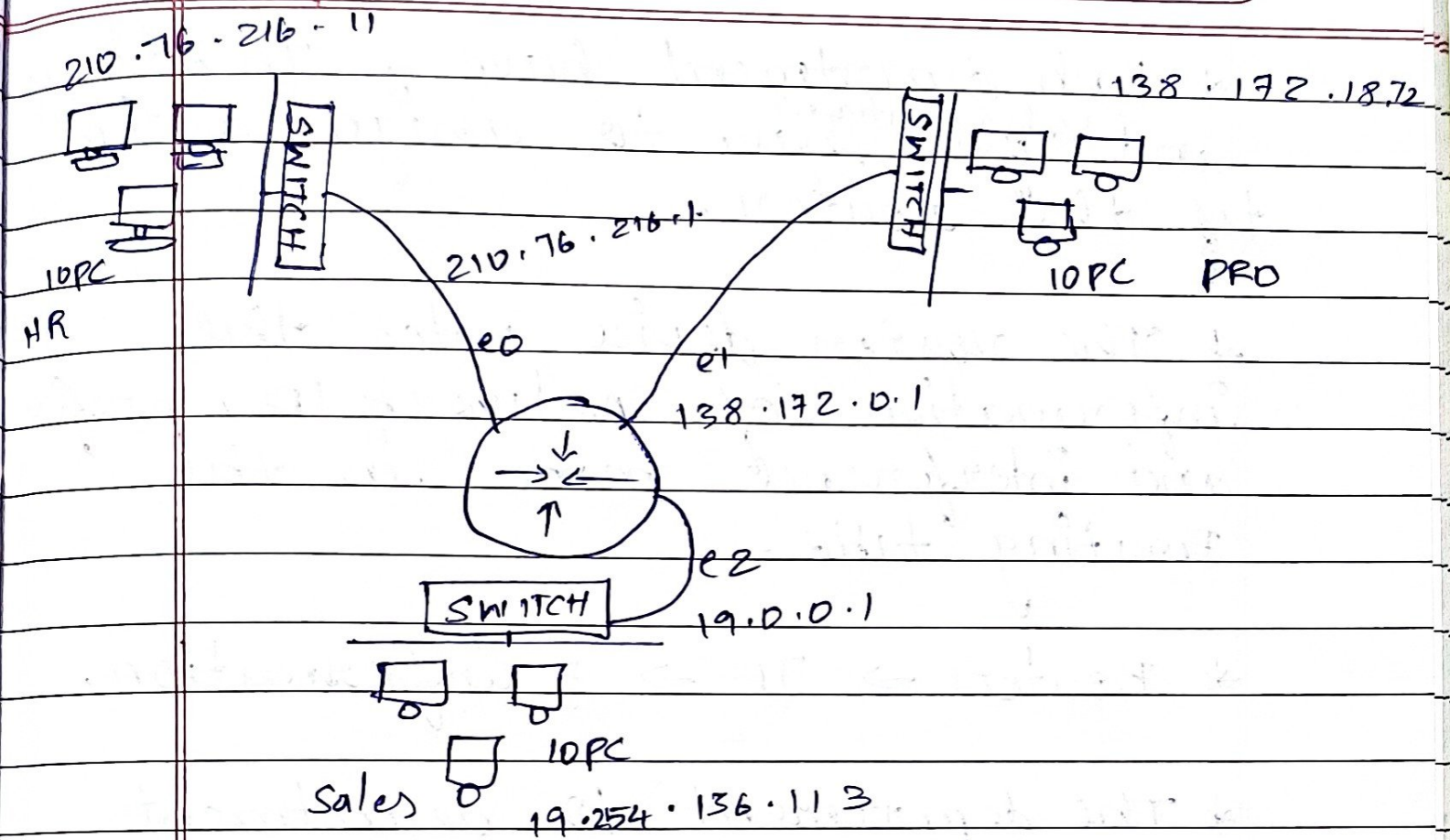
* This will be called as broadcast + unicast.

3) Explain behaviour of router device?

Router:-

- * It is used to break the broadcast domain.
- * It is a unicast device.
- * It contains routing table.
- * Routing table contains list of rules.

Let's see the below diagram to understand:



* you have three department as HR, production, sales.

* Each department pc has connected by the switch and all the switch can be connected by the Router device.

* The Router device performs the data sharing by the way of unicast.

* Each department have a IP Address and ~~Each department~~ have to assign a IP to the router.

* The router device notes the information of Network ID, prefix and interface name in the Routing table.

* Router $\rightarrow$ IP $\rightarrow$ 3 information.

* The department to department can be share the data using the router device by the way of unicast.

* This is called Router device.